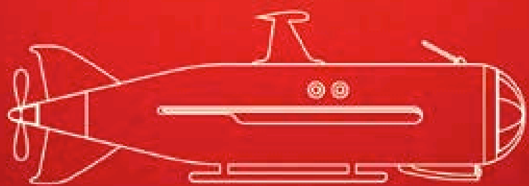
UNDER WATER

SENTRIES IN THE DEEP



The oceans of the world are some of the most remote and inaccessible places on the planet, and underwater drones are a perfect fit for working in this environment which is hostile for humans. From drilling oil rigs, to salvaging, to laying undersea cables, there is increasing undersea commercial activity around the world, and sophisticated robotic systems have already been developed for construction, repair, maintenance and monitoring applications.

These machines both work autonomously and with manual control. While the early versions were tethered, had limited range and battery capacities, the newer models are getting increasingly autonomous, with simple battery upgrades to existing platforms allowing for longer range and operation times. There are unmanned and untethered drones in the sea today for exploring seafloors, collecting samples, environmental monitoring and conducting oceanographic surveys. Such platforms lend themselves easily to military applications.

The US Navy pioneered the use of underwater drones in the 50s and the 60s, which were essentially cameras mounted on steerable craft for investigating shipwrecks, retrieving objects on the seafloor, disabling mines, and helping with the rescue operations of broken submarines. In 1985, a secret US Navy mission in collaboration with ocean explorer Robert Ballard to locate two wrecked nuclear submarines stumbled across the remains of the Titanic. This was a specially built remotely operated vehicle (ROV). Modern underwater drones were used for the search of the missing Malaysian Airlines Flight 370. The ROVs were deployed from dedicated research ships and were tethered to these boats with extremely long cables. The development of lithium ion batteries in the 2000s allowed these drones to go untethered, with artificial intelligence giving them automated capabilities. These drones require some level of autonomy as the conventional methods of navigation, such as using the GPS constellation, does not work in the underwater environment. Today, small consumer grade ROVs with cameras on board that can be controlled by a laptop are available on the market.

2003 saw the first deployment of military autonomous underwater vehicles (AUVs) as a part of Operation Iraqi Freedom, for the purpose of mine warfare. Currently the AUVs developed are for two primary purposes, anti-mine operations and intelligence, surveillance and reconnaissance (ISR) activities. These are autonomous underwater vehicles that can patrol the seas for long periods of time, looking for signs of enemy submarines. In the future though, they can be armed, used to deliver payloads or personnel, and select targets to neutralise on their own.

At the parade to mark the 70th anniversary of the founding of the People's Republic of China, a number of autonomous vehicles were displayed for the first time to the general public. Among these was the HSU-001, a large displacement AUV powered by twin propellers and loaded with sensors on retractable masts. All around the hull are thrusters that allow the drone six degrees of freedom while roaming the waters. From the design of the vehicle, it is apparent that it is meant for long term autonomous operations in the seas. The drone is comparable to the US Navy's Snakehead, which is capable of being launched from nuclear submarines, and scouting ahead for intelligence gathering activities. Dwarfing both of these is the Orca platform developed by Boeing, one of the versatile models being the Echo Voyager that is capable of being operated in both manned and unmanned configurations, deploying its own

Credit: Edgelab

L&T has developed three AUVs for DRDO in collaboration with Italian company Edgelab, known as Amogh, Adamya and Maya.